GFS - GlobalFoundries | Full investment artifact

Full investment research | Model assumptions and source evidence

Research update: October 11, 2026 | **Framework:** 8.0 | **Financial model:** TSMC interposer branch added; original product paths retained **Status:** NOT HELD in supplied original | **Classification:** BOTTLENECK CANDIDATE - differentiated photonics/SiGe within a diversified foundry **Artifact Base:** 40/60 = **67% - Watchlist** | **Portfolio Excel Base:** 67% (current October 7 owner setting retained; separate from artifact assessment) **Active price target:** Q4 2027 **USD 76.006228**, pricing Q4 2027-Q3 2028 at the retained 20x non-IFRS operating profit and 562M shares. Quarterly targets include the separately modeled interposer revenue and qualification costs below.

The forward opportunity is larger than the present CID revenue share suggests. Silicon photonics, SiGe and power can increase both market access and content per system. The full model carries that expansion into product GP while retaining the slower-growth foundry base, rising R&D and equity compensation allowance.

1. Supercycle, supply chain and durable advantage

The investment case is a mix shift toward differentiated optical, power and embedded technologies within a broad foundry. Photonics and SiGe are complementary, but neither implies that the entire company earns optical margins. New customer agreements support demand visibility; tool delivery, qualification and profitable capacity utilization determine the conversion to earnings.

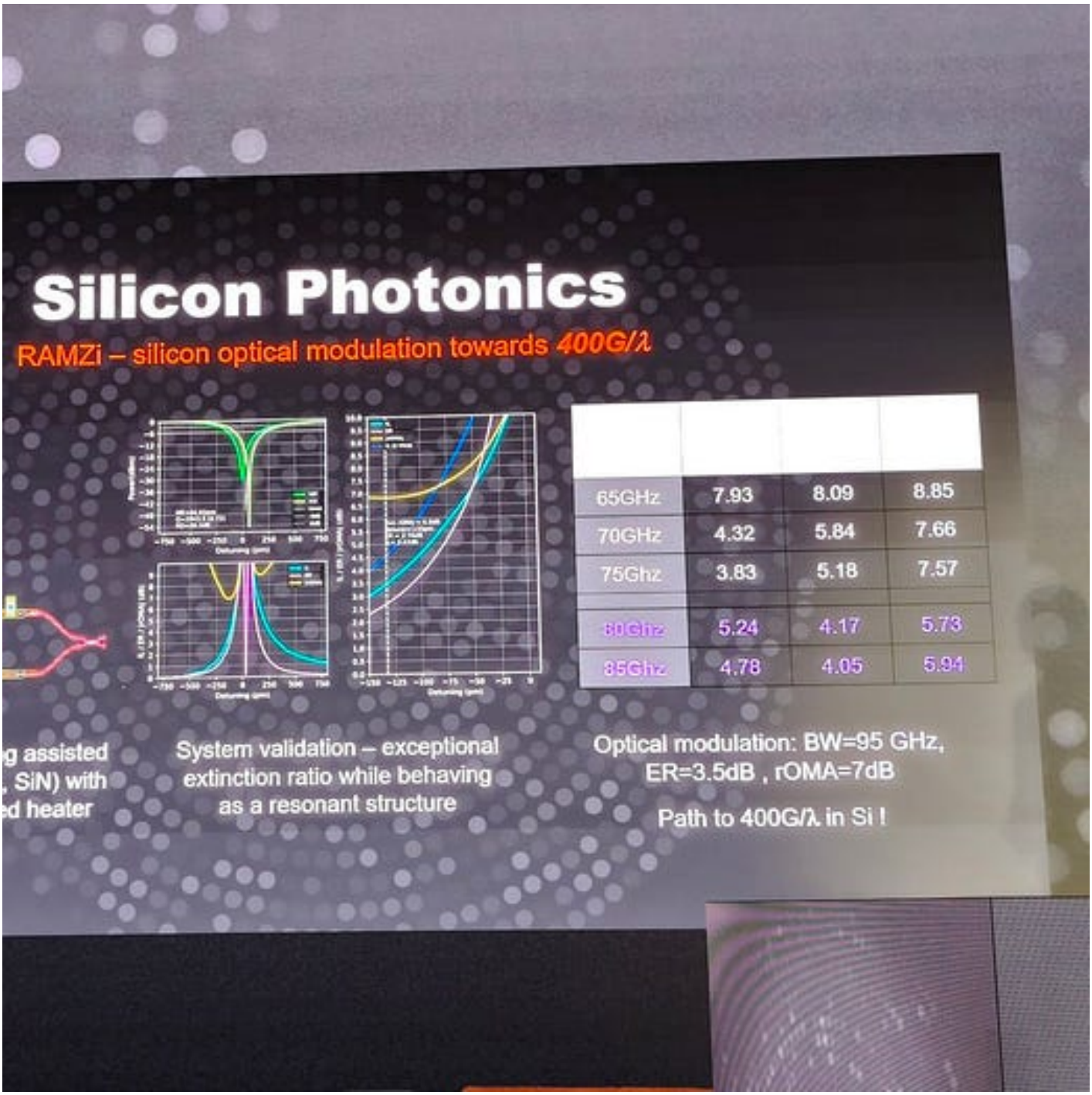
DB1 photonics and DB10 inference: optical bandwidth and electrical power demand → customer chip designs → qualified SiPho, SiGe and BCD processes → wafer fabrication and selected integration services. GF's process design kits, yield history and diversified footprint create qualification advantages. It is not a monopoly, and customers can use different foundry routes across products.

Business / curve	Stage	Evidence and next inflection
Pluggable SiPho and SiGe	Commercial ramp;10-50% qualitative	Current demand and signed capacity agreements support added qualified production.
NPO/CPO SCALE	Pre-ramp to early ramp	Tape-outs must pass qualification before recognized volume revenue.
Data-center power and physical AI	Early ramp	New processes and design partnerships broaden content; revenue timing varies.
Mobile/auto and legacy markets	Mixed mature/recovery	Not every end market shares the optical growth rate.

Stage percentages are qualitative framework bands relative to a possible mature opportunity, not measured market penetration. A current capacity constraint can coexist with a larger long-term TAM; qualification and economic capture decide how much reaches this company.

Engineering: compare an optical link, not a technology label

The February 12 Irrational Analysis memo makes a strong negative comparison between GlobalFoundries and Tower. Its useful engineering questions concern modulator bandwidth, optical loss, drive power, packaging and the qualified customer's complete link. Its language about an inferior process and inferred customer concentration goes beyond the evidence reproduced. A company can have successful SiGe, coherent-optics or silicon-photonics products without winning every CPO architecture. Conversely, a research demonstration does not establish production yield or commercial competitiveness. [E-GFS1]



Engineering exhibit tower-I12. Saved source: (Market Memo) Tower+GloFo Earnings, HBM4 Qualification Noise, 2026-02-12. Attribution and scope are discussed below.

RAMZI technical-slide photograph reproduced by the author. It presents device-level performance and a path toward 400G per wavelength. It is not a production-yield distribution, a customer allocation or a demonstrated 400G commercial module.

RAMZI combines interferometric and resonant elements. A fair comparison needs the same wavelength, electrical drive, optical input/output boundary, temperature range and receiver-error criterion. More bandwidth is useful only if the complete link retains adequate optical modulation amplitude, extinction ratio and noise margin at acceptable power and cost. Thermal control, packaging loss and manufacturing variation can alter the result. A small-signal bandwidth number alone cannot price a commercial wafer or determine the winning foundry.

The article assigns 90% confidence to an unnamed CPO win being Cisco and estimates that Cisco/Acacia represents 80-90% of GF's SiPho business. Those are source-authored estimates, not disclosed customer shares. The current model does not adopt those probabilities as measured allocations, nor infer that acquiring AMF or Infinilink proves failure of GF's existing platform. Process portfolio expansion, IP and engineering capability are alternative explanations; customer-level revenue and qualification evidence would distinguish them. [E-GFS1]

Current probability and cost treatment. Retain **70% conversion confidence** for the existing SCALE NPO/CPO programme, **85% execution confidence** for the Marvell SiGe agreement and **80% timing confidence** for MPS production. These are the existing ledger's confidence labels, not new haircuts or additive order values. The model already has SiPho sales of **$585M in 2027 and $900M in 2028**, with explicit product GMs of 52% and 55%; manufacturing depreciation stays in costs and R&D/opex rises in dollars. The February claims supply no newly identified lost customer, failed qualified process or price concession relative to the newer September evidence. Their incremental sales, COGS, opex and OP are zero; they do not justify a whole-company haircut or a transfer of revenue to Tower.

The engineering risk is nevertheless concrete. A design that needs more drive/laser power, more thermal correction, larger area or additional test/rework can have a lower accepted yield and higher cost per billed unit. The appropriate negative branch would isolate the exposed customer/process and recognition quarter, retain unavoidable fab and R&D costs, and net any substitution to another GF product. The appropriate positive branch would require a defensible increase in accepted units or net content after cannibalization. Neither a photographed eye diagram nor broad AI optical demand sizes that incremental invoice.

Newer Marvell and MPS agreements remain primary company evidence and are already inside other CID and the appropriate residual end markets. The model's $1.08B SiPho exit-2028 annualized rate is distinct from its $900M full-year sales. The engineering claims add no separate economics. With the October 11 interposer update, the **Q4 2027 price target is $76.006228**, retaining the existing 20x non-IFRS OP method and 562M share denominator. It does not change the current Excel Base of 67 or the separate analytical Base of 67.

Engineering sources, reviewed October 6, 2026: E-GFS1: Irrational Analysis, February 12, including the saved GF call excerpts and RAMZI photographs. G4: Marvell capacity partnership, September 17. G5: MPS partnership, September 9. The original technical-slide paper was not independently identified from the saved photograph, so its plotted results remain an attributed exhibit rather than a reproduced experiment.

GaN manufacturing: follow the wafer, then qualification

The May 17 Irrational Analysis article identifies GlobalFoundries as a manufacturing route behind Navitas's integrated GaN technology. Newer primary evidence provides a much clearer milestone: the September 1 joint announcement schedules first Gen5 GaNFast **wafers in September**, internal samples in October and strategic customer samples before year-end. The process uses GF's US 200mm GaN-on-silicon manufacturing platform. This verifies a named programme and a staged manufacturing route; it does not say that qualified AI power systems are already shipping in volume. [E-GFS2, E-GFS3]

The value split matters. Navitas sells the power devices and integrated functionality; GF earns the applicable foundry invoice. End-system watts, a transistor's maximum heat dissipation and Navitas device revenue are not interchangeable with GF wafer sales. Better integration can reduce customer board area or protection

effort, but GF must convert process qualification into good wafers, customer acceptance and profitable utilization. Startup scrap, added process steps, test and depreciation can offset a richer selling price.

Probability and bounded treatment. The announced schedule is a verified statement, while execution remains uncertain. Assign **75% analyst confidence** to strategic customer sampling by the end of 2026. For financial recognition, use a bounded diagnostic of **0-1% of the existing $1,150M CY2027 other-CID forecast**, or **$0-11.5M** of possible Navitas-related foundry revenue within that category. The zero lower bound reflects sample-to-volume timing; the upper bound is a deliberately small attribution proxy, not a disclosed wafer commitment. Probability weighting gives $0-8.625M before the **100% overlap adjustment** against the current power/communications growth category.

At the category's existing 40% product-GM assumption, the conditional range implies up to **$6.9M of manufacturing cost and $4.6M of GP**, before allocated development/customer-support expense. Early programme margin may be lower; those costs cannot be omitted to turn a sampling announcement into pure profit. The current model's other-CID trajectory already includes data-center power alongside SiGe and communications, and the consolidated opex path rises from $1,180M in 2027 to $1,380M in 2028. Incremental revenue, COGS, opex and OP remain **zero** after overlap. The new evidence identifies a plausible constituent of that demanding growth path rather than enlarging it without a wafer/price bridge.

This treatment is not a requirement to wait for official revenue guidance. A credible customer wafer schedule, qualified usable capacity, billable pricing or newly identified supply restriction could change the active forecast before reported revenue. The immediate checks are October internal samples, year-end customer samples and conversion into accepted production wafers. Neither the announcement nor the older article establishes exclusive supply, profitable high-volume yield or a separately additive contract value.

Q4 2027 is $76.006228 after the separate interposer update, using the retained 20x forward non-IFRS OP method, 562M share denominator, current Excel Base67 and analytical Base67. Existing SCALE70%, Marvell85% and MPS80% confidence labels are not applied again to the same sales.

Sources: E-GFS2: Irrational Analysis, May 17, 2026, including its integrated-driver and GaN manufacturing discussion; E-GFS3: GF/Navitas Gen5 US manufacturing milestone, September 1, 2026. The source timetable distinguishes first wafers, internal samples and customer samples; later volume recognition remains a model assumption.

October 11: TSMC interposer agreement

TSMC interposers: a new invoice, with a later factory ramp

GlobalFoundries announced on October 8 a $2 billion manufacturing agreement with TSMC, initially lasting five years. GF plans additional Malta capacity for silicon interposers supporting CoWoS, including embedded deep-trench capacitor components, with volume production expected to begin ramping in the first half of 2028. The disclosed agreement is a verified event; its timing and profitable delivery remain forward-looking. This manufacturing service is distinct from selling a photonic wafer or an entire AI package.

The existing model already includes selective 2027 pricing, the Marvell SiGe relationship, MPS and a substantial SiPho/other-CID expansion. The TSMC interposer contract was absent from that earlier source register, but generic other-CID growth can overlap the new business. We therefore reserve 50% of the conditional programme revenue against that existing growth allowance, leaving SiPho forecasts unchanged. This is an explicit model overlap assumption, not a disclosed customer allocation.

The programme's existence is established. For execution, the model uses 85% probability on a bounded accepted-sales schedule: $200 million gross programme sales in 2028, within a $100-300 million conditional range, followed by $400 million in each of 2029 and 2030. The 2028 profile is $0/$25/$75/$100 million, starting in Q2

within the issuer's first-half ramp window. Subsequent quarters are $75/$100/$100/$125 million in 2029 and $100 million each in 2030. Those annual and quarterly allocations are analytical assumptions, not contracted minimum purchases. The gross model schedule totals $1 billion through 2030; the undistributed remainder of the headline agreement is not accelerated into the current forecast or treated as perpetual sales.

After the 85% execution weight and 50% overlap reserve, additional sales are $85 million in 2028 and $170 million in each of 2029 and 2030. The programme gross-margin assumptions are 30% at ramp and 35% subsequently, including manufacturing depreciation, process yield, scrap and startup factory costs. They are below the existing other-CID margins of 43%/45%/46%. A further 5% of incremental revenue funds engineering and customer support, producing $21.25 million incremental operating profit in 2028 and $51 million in each of 2029 and 2030.

Qualification must be paid before successful sales. We use a $10 million conditional engineering budget, 0.5% of the disclosed agreement amount, with 50% already covered by existing operating expenses. The remaining $5 million is expensed $2.5 million in each of Q3 and Q4 2027. Those expected incurred costs are not discounted by the probability that production ultimately succeeds. Manufacturing startup costs remain in the product-margin assumption, avoiding a second charge for the same factory activity.

The active Q4 2027 window captures the Q4 2027 qualification cost and the first three quarters of 2028, not the full 2028 contract contribution. Its incremental forward operating profit is $8.125 million. The retained method remains 20 times non-IFRS operating profit, including manufacturing depreciation but excluding recurring SBC, divided by the target quarter's existing 562 million diluted-share proxy. No cash, debt, multiple, share-schedule or score change is introduced. Excel Base67 and analytical Base67 are retained; old artifact wording describing a protected Excel Base63 predates the owner's October7 override removal.

The primary announcement does not disclose minimum purchase enforceability, annual orders, payment composition, capex, usable wafer capacity, customer advances, subsidy terms or a product-margin bridge. The numerical case uses bounded proxies for those incomplete operating details rather than assuming that all $2 billion is near-term, pure-margin revenue. Material dedicated capex, poor yield, qualification delays or displacement of more existing production than the overlap reserve would reduce the outcome. The next useful evidence is Malta tool installation and customer qualification, the accepted shipment schedule, and the relationship between manufacturing invoices, funding and depreciation.

Source: GlobalFoundries, October 8, 2026, https://gf.com/news-and-events/news/globalfoundries-reaches-agreement-to-establish-us-based-supply-of-silicon-interposers-for-advanced-ai-packaging/

Quarter	Conditional sales	Net added sales	Net added GP	Extra opex	Added OP
Q3 2027	0.000	0.000	0.000	2.500	-2.500
Q4 2027	0.000	0.000	0.000	2.500	-2.500
Q1 2028	0.000	0.000	0.000	0.000	0.000
Q2 2028	25.000	10.625	3.188	0.531	2.656
Q3 2028	75.000	31.875	9.562	1.594	7.969
Q4 2028	100.000	42.500	12.750	2.125	10.625
Q1 2029	75.000	31.875	11.156	1.594	9.562
Q2 2029	100.000	42.500	14.875	2.125	12.750

Quarter	Conditional sales	Net added sales	Net added GP	Extra opex	Added OP
Q3 2029	100.000	42.500	14.875	2.125	12.750
Q4 2029	125.000	53.125	18.594	2.656	15.938
Q1 2030	100.000	42.500	14.875	2.125	12.750
Q2 2030	100.000	42.500	14.875	2.125	12.750
Q3 2030	100.000	42.500	14.875	2.125	12.750
Q4 2030	100.000	42.500	14.875	2.125	12.750

The $2B headline is not recognized as revenue at signing. The modeled 2028-30 gross sales sum $1B before probability and overlap. The remainder is not automatically booked, because the initial term dates and payment mix are undisclosed. A capital-budget diagnostic of 20-40% of the headline value is $400-800M before any customer funding, reuse or subsidy. This is a financing stress range, not a new cash-flow forecast. Manufacturing depreciation and startup loss are already charged through the branch gross margins. No debt or share change is asserted without financing evidence.

2. Forward TAM and the three-phase expansion lens

Phase	Engineering opportunity	Demand-to-company bridge
Phase 1;100% thematic	Current optical links and differentiated foundry work	More links, higher lane speeds and new designs expand the served wallet. Wafer content depends on die area, integration and yield rather than a fixed fraction of optical-module ASP.
Phase 2;50% thematic	NPO/CPO, SiGe and data-center power	SiPho target framing extends from >1B exit 2028 toward 2B 2030 run rate. The model uses 900M/1,950M actual annual sales equivalents and 1.08B/2.16B exit annualized values. Marvell and MPS agreements support availability/capture without disclosed additive contract values.
Phase 3;20% thematic	Quantum, physical AI, embedded compute and wearables	New IP and process combinations can create design access before wafer revenue. Government R&D funding is milestone-based support, not immediate commercial revenue; services already sit inside reported end markets.

The 100% / 50% / 20% thematic weights guide thesis assessment. They are not blanket discounts to reported sales, contracts or all forecasts. Programme probabilities below govern only separately identified optionality. Current reported sales calibrate the starting point; they do not cap the future market.

3. Current earnings and management evidence

Q2 2026 revenue was USD 1,786M. Non-IFRS GP/OP were 534M/298M, versus IFRS 505M/174M. Q3 guidance midpoint is revenue 1,885M, non-IFRS GM 30.5% and opex 260M, implying OP 314.925M. The guided fully diluted count is 556M, replacing the old 548.42M denominator.

Reviewed the latest August 5 earnings release, substantive Q2 call passages and September 8 Goldman Sachs discussion, with a Q1 transcript cross-check. Complete sequential coverage of both earnings-call Q&As was not achieved. September 9 and September 17 agreements are incorporated. The September discussion

reinforces supply/qualification constraints rather than a new quantified revenue guide.

Latest financial release, Q2 management remarks/Q&A passages, Q1 cross-check and the newer Goldman discussion were reviewed. Sequential coverage of both complete earnings calls is not claimed. Product margins, wafer-equivalent ASP and outer-year capture remain analytical assumptions, even where the direction is supported by signed partnerships.

4. Local evidence and calibration packet

The latest issuer release anchors Q2 actuals and the Q3 midpoint. The forward series starts in Q3 and uses estimated product allocations 85M SiPho+225M other CID+1,575M other=1,885M. Product GP 42.5+85.5+446.925=574.925M, exactly 30.5%; subtract 260M expenses for 314.925M OP. Initial product GM calibration is not evidence of disclosed segment costs. Q2 IFRS OP 174M versus adjusted 298M illustrates a 124M basis difference.

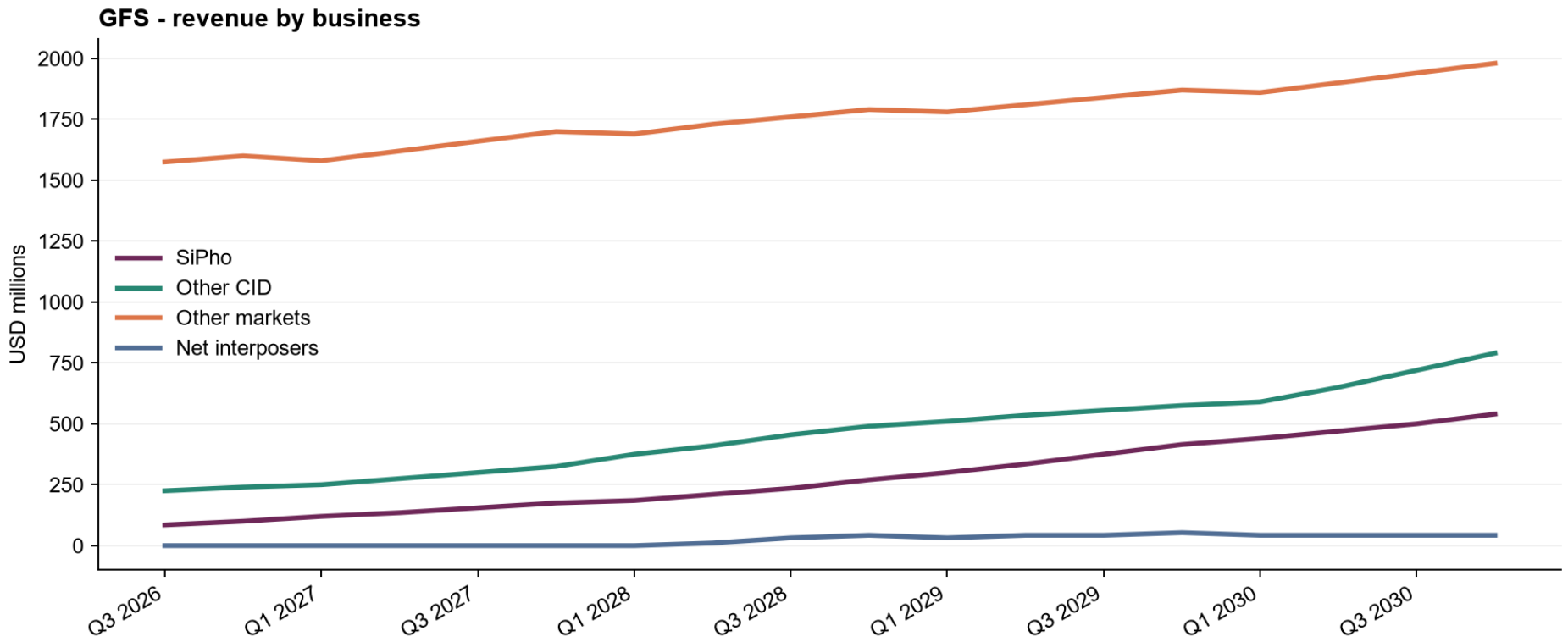
5. Summing thesis score

Category	Points / maximum	Evidence
Core Thesis	16/20	Verified optical capacity demand and customer commitment; competition and legacy exposure remain.
Cycle	7/10	Optical ramp offsets mobile weakness; capacity installation gates conversion.
TAM	8/10	NPO/CPO, data-center power and physical AI broaden available content.
Catalysts	5/5	Marvell capacity agreement, MPS production in 2027 and SCALE qualifications.
Macro	7/10	AI demand and diversified manufacturing benefit; handset and auto cycles still matter.
Alpha	2/5	Differentiation is increasingly visible and publicly promoted.
Risks	-5	Capex, customer concentration, ramp/yield, competition, recurring SBC and geopolitical risk.

Sum: 40/60 × 100 = 66.666667%, conventionally rounded to 67%. This advisory thesis score does not replace any portfolio Base override. The separate risk classes below total the displayed deduction; holding status and classification do not waive that deduction.

Financial views of the current model

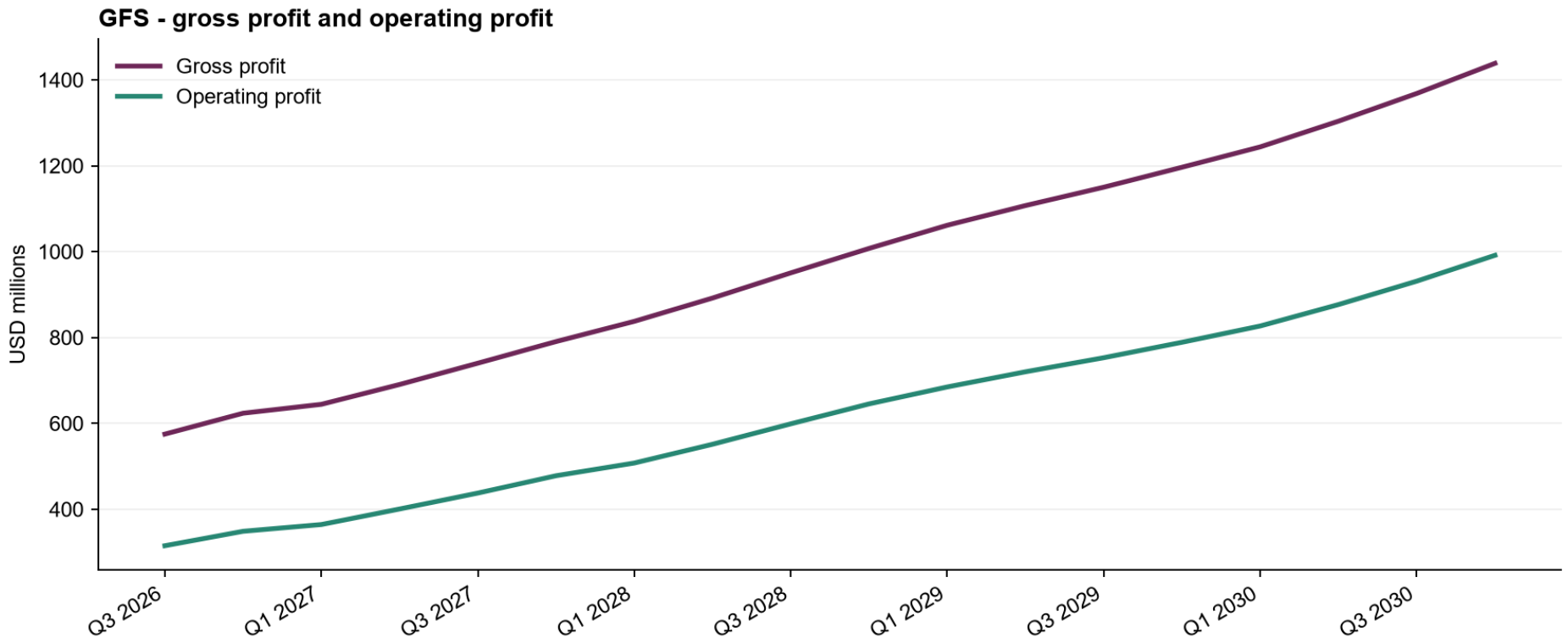
These views include the October 11 interposer branch and retain the valuation convention. Financial values are USD millions except per-share ceilings and percentages. Gold identifies the active Q4 2027 point on the first series where shown; the separately described interposer branch is the forecast change. Underlying figures are reproduced below each chart.



GFS - revenue by business

Quarter	Silicon photonics	Other CID	Other markets	Interposer increment
Q3 2026	85.000	225.000	1,575.000	0.000
Q4 2026	100.000	240.000	1,600.000	0.000
Q1 2027	120.000	250.000	1,580.000	0.000
Q2 2027	135.000	275.000	1,620.000	0.000
Q3 2027	155.000	300.000	1,660.000	0.000
Q4 2027	175.000	325.000	1,700.000	0.000
Q1 2028	185.000	375.000	1,690.000	0.000
Q2 2028	210.000	410.000	1,730.000	10.625

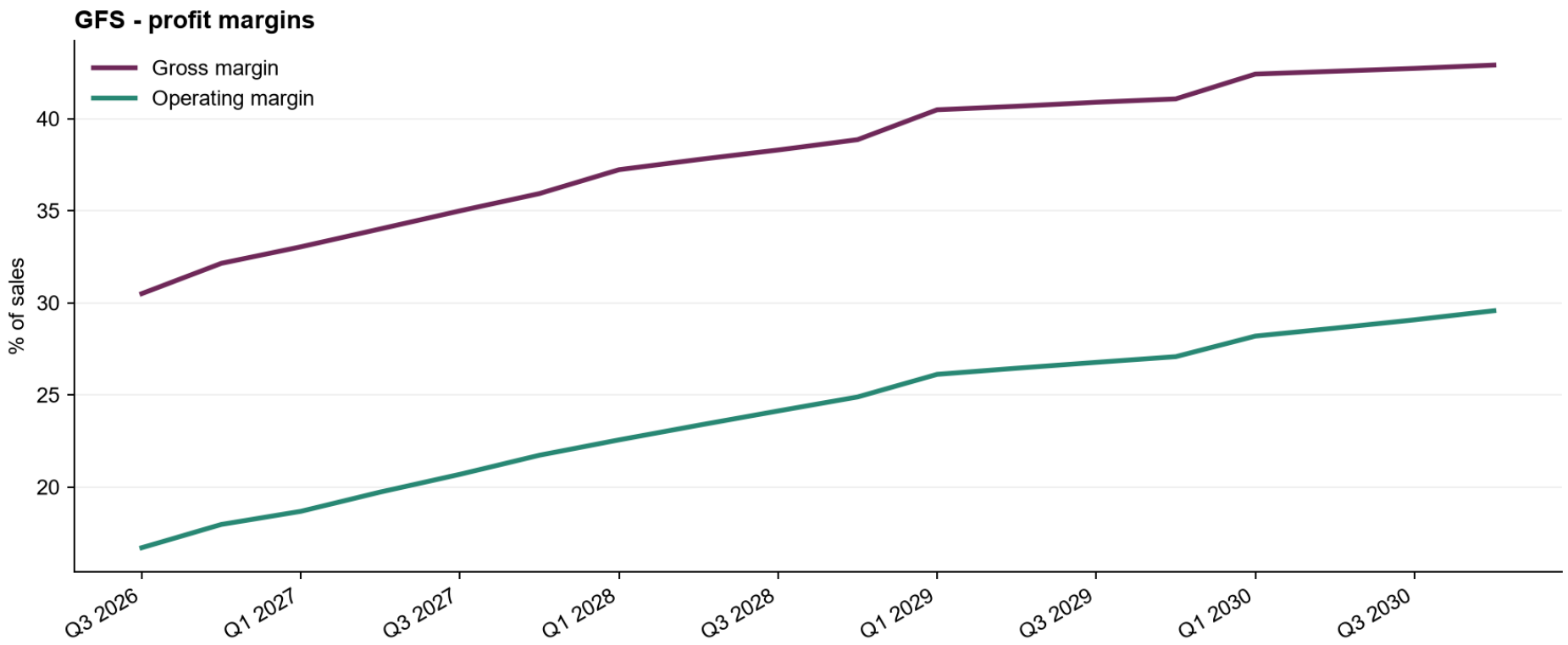
Quarter	Silicon photonics	Other CID	Other markets	Interposer increment
Q3 2028	235.000	455.000	1,760.000	31.875
Q4 2028	270.000	490.000	1,790.000	42.500
Q1 2029	300.000	510.000	1,780.000	31.875
Q2 2029	335.000	535.000	1,810.000	42.500
Q3 2029	375.000	555.000	1,840.000	42.500
Q4 2029	415.000	575.000	1,870.000	53.125
Q1 2030	440.000	590.000	1,860.000	42.500
Q2 2030	470.000	650.000	1,900.000	42.500
Q3 2030	500.000	720.000	1,940.000	42.500
Q4 2030	540.000	790.000	1,980.000	42.500



GFS - gross profit and operating profit

Quarter	Gross profit	Operating profit
Q3 2026	574.925	314.925
Q4 2026	623.600	348.600
Q1 2027	644.300	364.300
Q2 2027	690.500	400.500
Q3 2027	740.100	437.600
Q4 2027	790.500	478.000
Q1 2028	837.600	507.600
Q2 2028	891.838	551.306

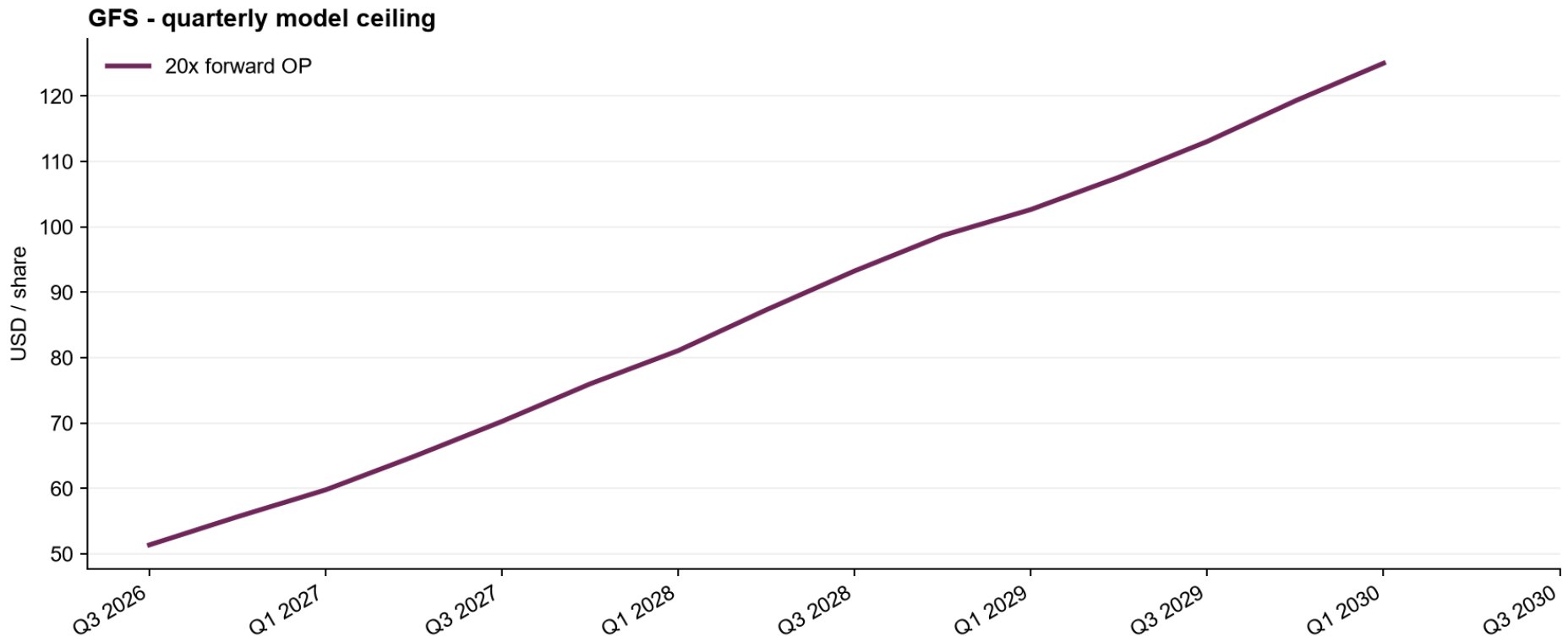
Quarter	Gross profit	Operating profit
Q3 2028	950.462	598.869
Q4 2028	1,007.400	645.275
Q1 2029	1,061.356	684.762
Q2 2029	1,107.225	720.100
Q3 2029	1,149.975	752.850
Q4 2029	1,196.444	788.787
Q1 2030	1,243.875	826.750
Q2 2030	1,303.775	876.650
Q3 2030	1,368.275	931.150
Q4 2030	1,438.475	991.350



GFS - profit margins

Quarter	Gross margin	Operating margin
Q3 2026	30.500	16.707
Q4 2026	32.144	17.969
Q1 2027	33.041	18.682
Q2 2027	34.015	19.729
Q3 2027	34.993	20.690
Q4 2027	35.932	21.727
Q1 2028	37.227	22.560
Q2 2028	37.780	23.354

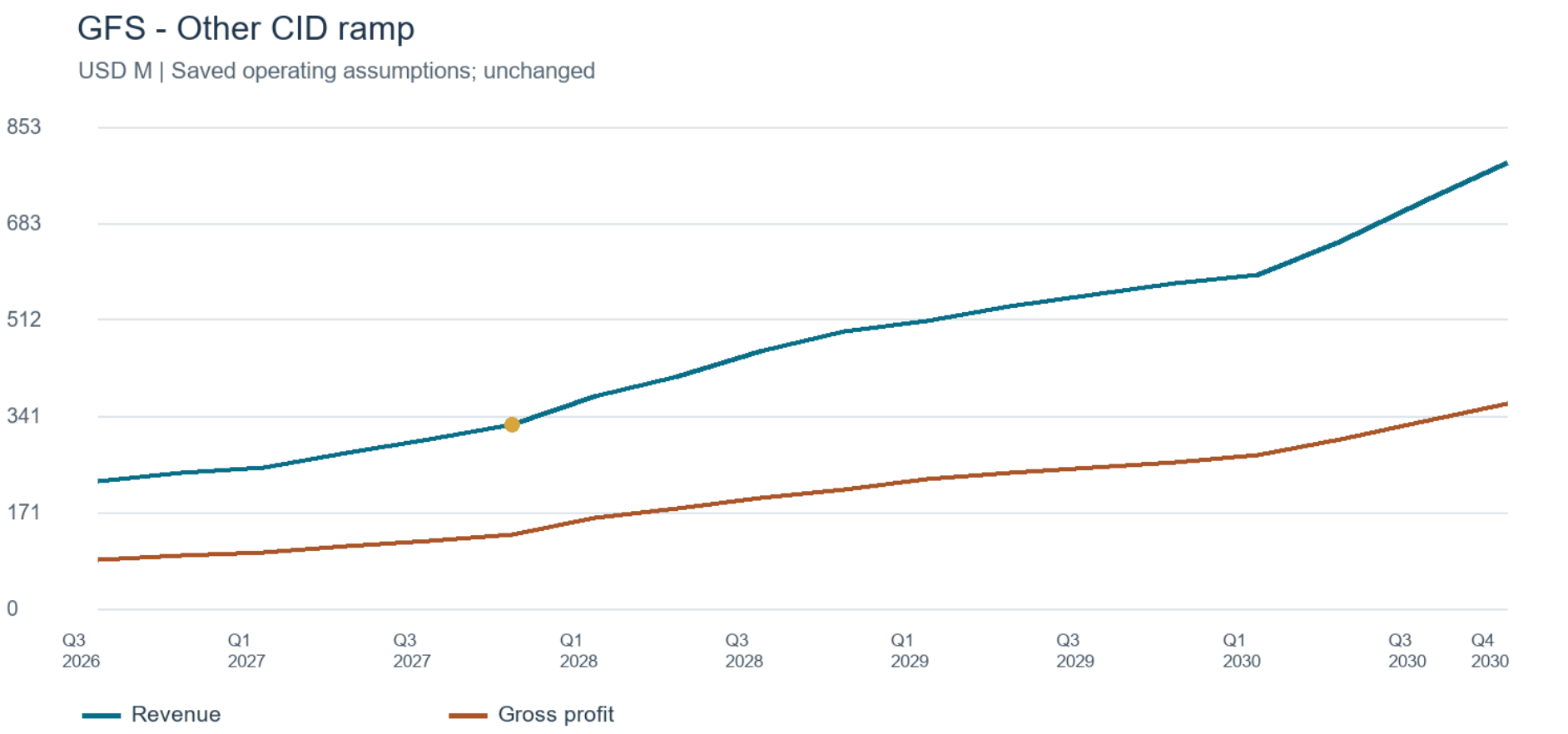
Quarter	Gross margin	Operating margin
Q3 2028	38.296	24.130
Q4 2028	38.858	24.890
Q1 2029	40.481	26.117
Q2 2029	40.669	26.450
Q3 2029	40.888	26.768
Q4 2029	41.071	27.077
Q1 2030	42.417	28.193
Q2 2030	42.572	28.625
Q3 2030	42.725	29.076
Q4 2030	42.908	29.570



GFS - quarterly model ceiling

Quarter	Forward OP valuation
Q3 2026	51.378597
Q4 2026	55.691203
Q1 2027	59.800712
Q2 2027	64.900356
Q3 2027	70.267126
Q4 2027	76.006228
Q1 2028	81.093310
Q2 2028	87.331426

Quarter	Forward OP valuation
Q3 2028	93.274868
Q4 2028	98.696743
Q1 2029	102.665505
Q2 2029	107.612805
Q3 2029	113.067509
Q4 2029	119.280052
Q1 2030	125.031034



GFS - Other CID ramp

Quarter	Revenue	Gross profit
Q3 2026	225.000	85.500
Q4 2026	240.000	93.600
Q1 2027	250.000	100.000
Q2 2027	275.000	110.000
Q3 2027	300.000	120.000

Quarter	Revenue	Gross profit
Q4 2027	325.000	130.000
Q1 2028	375.000	161.250
Q2 2028	410.000	176.300
Q3 2028	455.000	195.650
Q4 2028	490.000	210.700
Q1 2029	510.000	229.500
Q2 2029	535.000	240.750
Q3 2029	555.000	249.750
Q4 2029	575.000	258.750
Q1 2030	590.000	271.400
Q2 2030	650.000	299.000
Q3 2030	720.000	331.200
Q4 2030	790.000	363.400

6. Revenue sources, capture and product margins

Calendar quarters; preserve 20x non-IFRS OP, not EBITDA. Separate SiPho, other CID (including SiGe and data-center power), and all other markets/services. These mutually exclusive buckets avoid adding IP/services a second time. Q3 product revenue is an analyst split of the consolidated midpoint, not disclosed segment guidance. Its product GMs reconcile exactly to 30.5% consolidated GM and USD 260M opex. Subsequent quarterly revenue, mix and margins are LOW-confidence price target assumptions. SiPho reaches USD 1.08B annualized in Q4 2028 (actual model FY2028 sales USD 900M) and USD 2.16B annualized in Q4 2030 (FY2030 USD 1.95B). Other CID includes the existing Marvell relationship and MPS-related data-center power; the rest of MPS end-market exposure remains in other markets. No additional agreement-value layer is added. Legacy margins recover with utilization and mix, while R&D/opex rises in dollars. Displayed product GMs are analytical allocations, not company disclosures. Shares start at guided 556M, then rise to 562/568/574/580M for 2027-30 to allow for continuing equity compensation; no unexecuted buyback benefit is assumed. Grants are not added to OP or revenue. Full future option dilution is not independently reconstructed.

Vector	Actual calibration / source	Forward demand and capture	Margin derivation / constraint
Silicon photonics	85M Q3 allocation; company discloses a broader CID segment.	FY27 sales 585M, FY28 900M, FY29 1,425M, FY30 1,950M. FY30/FY27=3.333333×. An illustrative 2.5×productive wafer-equivalent output ×1.333333 die/process content provides the bridge; neither factor is disclosed capacity guidance. The module-to-wafer capture is not inferred from total optical TAM alone.	52% 2027→57% 2029-30 non-IFRS product GM proxy. Implied product cost falls 48→43% with loading/yield and differentiated mix; manufacturing depreciation remains included.
Other CID	225M Q3 allocation includes SiGe, power and communications.	Annual 1,150/1,730/2,175/2,750M. Marvell capacity and MPS power are within these totals. At an illustrative 3,000 USD per recognized wafer-equivalent invoice,2030 requiresabout 916,667 equivalents; this is a sensitivity denominator, not a disclosed shipment forecast.	40% 2027→46% 2030 proxy. Increased content may also raise mask/process cost, so revenue growth alone cannot establish the margin.
Other markets/services	1,575M Q3 residual.	Annual 6,560/6,970/7,300/7,680M. Growth reflects a gradual recovery and richer automotive/IoT/IP mix, not optical growth applied to the whole company. Technology services are already included.	2027 GM 30.5→33.5%,2028 34→35.5%,2029 36.5%,2030 38%. These are demanding mix/productivity assumptions, not current broad-market guidance.

Confidence: reported historical totals are HIGH. Source-based forward extrapolations are MEDIUM only where the cited source sizes the relevant period; other product allocations, future capture and margin assumptions are LOW. A source for the historical anchor does not make an unsourced future factor management guidance. The detailed factors below are the single modeled base case, with uncertain assumptions exposed for review.

V1. Silicon photonics (within CID)

Sources: G1/2/3/7; product allocation and exit-run-rate trajectory. Revenue = anchor × factor; GP/contribution = revenue × displayed product margin. Forecast confidence: LOW unless a specific actual/guided total is identified above.

Quarter	Revenue anchor	Demand / conversion factor	Revenue	Product margin	GP / contribution
Q3 2026	85.000000	1.000000	85.000	50.000%	42.500
Q4 2026	85.000000	1.176471	100.000	50.000%	50.000

Quarter	Revenue anchor	Demand / conversion factor	Revenue	Product margin	GP / contribution
Q1 2027	85.000000	1.411765	120.000	52.000%	62.400
Q2 2027	85.000000	1.588235	135.000	52.000%	70.200
Q3 2027	85.000000	1.823529	155.000	52.000%	80.600
Q4 2027	85.000000	2.058824	175.000	52.000%	91.000
Q1 2028	85.000000	2.176471	185.000	55.000%	101.750
Q2 2028	85.000000	2.470588	210.000	55.000%	115.500
Q3 2028	85.000000	2.764706	235.000	55.000%	129.250
Q4 2028	85.000000	3.176471	270.000	55.000%	148.500
Q1 2029	85.000000	3.529412	300.000	57.000%	171.000
Q2 2029	85.000000	3.941176	335.000	57.000%	190.950
Q3 2029	85.000000	4.411765	375.000	57.000%	213.750
Q4 2029	85.000000	4.882353	415.000	57.000%	236.550
Q1 2030	85.000000	5.176471	440.000	57.000%	250.800
Q2 2030	85.000000	5.529412	470.000	57.000%	267.900
Q3 2030	85.000000	5.882353	500.000	57.000%	285.000
Q4 2030	85.000000	6.352941	540.000	57.000%	307.800

V2. Other CID: SiGe, power and communications

Sources: G2/4/5; SiGe and power demand, undisclosed contract values. Revenue = anchor × factor; GP/contribution = revenue × displayed product margin. Forecast confidence: LOW unless a specific actual/guided total is identified above.

Quarter	Revenue anchor	Demand / conversion factor	Revenue	Product margin	GP / contribution
Q3 2026	225.000000	1.000000	225.000	38.000%	85.500
Q4 2026	225.000000	1.066667	240.000	39.000%	93.600
Q1 2027	225.000000	1.111111	250.000	40.000%	100.000

Quarter	Revenue anchor	Demand / conversion factor	Revenue	Product margin	GP / contribution
Q2 2027	225.000000	1.222222	275.000	40.000%	110.000
Q3 2027	225.000000	1.333333	300.000	40.000%	120.000
Q4 2027	225.000000	1.444444	325.000	40.000%	130.000
Q1 2028	225.000000	1.666667	375.000	43.000%	161.250
Q2 2028	225.000000	1.822222	410.000	43.000%	176.300
Q3 2028	225.000000	2.022222	455.000	43.000%	195.650
Q4 2028	225.000000	2.177778	490.000	43.000%	210.700
Q1 2029	225.000000	2.266667	510.000	45.000%	229.500
Q2 2029	225.000000	2.377778	535.000	45.000%	240.750
Q3 2029	225.000000	2.466667	555.000	45.000%	249.750
Q4 2029	225.000000	2.555556	575.000	45.000%	258.750
Q1 2030	225.000000	2.622222	590.000	46.000%	271.400
Q2 2030	225.000000	2.888889	650.000	46.000%	299.000
Q3 2030	225.000000	3.200000	720.000	46.000%	331.200
Q4 2030	225.000000	3.511111	790.000	46.000%	363.400

V3. Other markets and services, net of CID

Sources: G1/2/7; remaining end markets including technology services. Revenue = anchor × factor; GP/contribution = revenue × displayed product margin. Forecast confidence: LOW unless a specific actual/guided total is identified above.

Quarter	Revenue anchor	Demand / conversion factor	Revenue	Product margin	GP / contribution
Q3 2026	1,575.000000	1.000000	1,575.000	28.376%	446.925
Q4 2026	1,575.000000	1.015873	1,600.000	30.000%	480.000
Q1 2027	1,575.000000	1.003175	1,580.000	30.500%	481.900
Q2 2027	1,575.000000	1.028571	1,620.000	31.500%	510.300

Quarter	Revenue anchor	Demand / conversion factor	Revenue	Product margin	GP / contribution
Q3 2027	1,575.000000	1.053968	1,660.000	32.500%	539.500
Q4 2027	1,575.000000	1.079365	1,700.000	33.500%	569.500
Q1 2028	1,575.000000	1.073016	1,690.000	34.000%	574.600
Q2 2028	1,575.000000	1.098413	1,730.000	34.500%	596.850
Q3 2028	1,575.000000	1.117460	1,760.000	35.000%	616.000
Q4 2028	1,575.000000	1.136508	1,790.000	35.500%	635.450
Q1 2029	1,575.000000	1.130159	1,780.000	36.500%	649.700
Q2 2029	1,575.000000	1.149206	1,810.000	36.500%	660.650
Q3 2029	1,575.000000	1.168254	1,840.000	36.500%	671.600
Q4 2029	1,575.000000	1.187302	1,870.000	36.500%	682.550
Q1 2030	1,575.000000	1.180952	1,860.000	38.000%	706.800
Q2 2030	1,575.000000	1.206349	1,900.000	38.000%	722.000
Q3 2030	1,575.000000	1.231746	1,940.000	38.000%	737.200
Q4 2030	1,575.000000	1.257143	1,980.000	38.000%	752.400

7. Operating-cost bridge and quarterly operating model

The explicit dollar expense schedule begins 260M in Q3 and 275M in Q4, with subsequent R&D and selling costs rising through the model. Recent IP/design-team acquisitions require spending before the related wafer revenues mature. Product GP already includes manufacturing depreciation; no EBITDA add-back enters valuation. Non-IFRS measures exclude recurring SBC. The Q3 guide assumes roughly 76M SBC, an economic compensation cost that a share allowance does not erase.

Profit basis: Non-IFRS operating profit, including manufacturing depreciation. Amounts below are USD millions.

Quarter	Status	Revenue	GP / contribution	Opex %	Opex	OP margin	Operating profit	Main driver	Confidence
Q3 2026	Model (guide midpoint)	1,885.000	574.925	13.793%	260.000	16.707%	314.925	Q3 revenue/GM/expense midpoint	LOW model

Quarter	Status	Revenue	GP / contribution	Opex %	Opex	OP margin	Operating profit	Main driver	Confidence
Q4 2026	Model	1,940.000	623.600	14.175%	275.000	17.969%	348.600	Quarterly shipment recovery	LOW model
Q1 2027	Model	1,950.000	644.300	14.359%	280.000	18.682%	364.300	2027 pricing and MPS launch	LOW model
Q2 2027	Model	2,030.000	690.500	14.286%	290.000	19.729%	400.500	NPO qualification conversion	LOW model
Q3 2027	Model	2,115.000	740.100	14.303%	302.500	20.690%	437.600	Higher optical content; interposer qualification cost	LOW model
Q4 2027	Model	2,200.000	790.500	14.205%	312.500	21.727%	478.000	2027 qualified capacity; interposer qualification cost	LOW model
Q1 2028	Model	2,250.000	837.600	14.667%	330.000	22.560%	507.600	2028 SiPho ramp; pre-volume interposer phase	LOW model
Q2 2028	Model	2,360.625	891.838	14.425%	340.531	23.354%	551.306	Optical yield/loading; interposer sales after overlap	LOW model
Q3 2028	Model	2,481.875	950.462	14.166%	351.594	24.130%	598.869	CPO adoption scenario; interposer sales after overlap	LOW model

Quarter	Status	Revenue	GP / contribution	Opex %	Opex	OP margin	Operating profit	Main driver	Confidence
Q4 2028	Model	2,592.500	1,007.400	13.968%	362.125	24.890%	645.275	SiPho 1.08B exit annualized; interposer sales after overlap	LOW model
Q1 2029	Model	2,621.875	1,061.356	14.364%	376.594	26.117%	684.762	2029 broader optical demand; interposer sales after overlap	LOW model
Q2 2029	Model	2,722.500	1,107.225	14.219%	387.125	26.450%	720.100	More qualified output; interposer sales after overlap	LOW model
Q3 2029	Model	2,812.500	1,149.975	14.120%	397.125	26.768%	752.850	Power and photonics mix; interposer sales after overlap	LOW model
Q4 2029	Model	2,913.125	1,196.444	13.994%	407.656	27.077%	788.787	2029 customer scale; interposer sales after overlap	LOW model
Q1 2030	Model	2,932.500	1,243.875	14.224%	417.125	28.193%	826.750	2030 SiPho progression; interposer sales after overlap	LOW model
Q2 2030	Model	3,062.500	1,303.775	13.947%	427.125	28.625%	876.650	Capacity conversion; interposer sales after overlap	LOW model

Quarter	Status	Revenue	GP / contribution	Opex %	Opex	OP margin	Operating profit	Main driver	Confidence
Q3 2030	Model	3,202.500	1,368.275	13.649%	437.125	29.076%	931.150	Higher optical penetration; interposer sales after overlap	LOW model
Q4 2030	Model	3,352.500	1,438.475	13.337%	447.125	29.570%	991.350	SiPho 2.16B exit annualized; interposer sales after overlap	LOW model

OP = sum of product GP/contribution - explicit operating costs. OP margin = OP ÷ revenue. Displayed numbers are rounded for reading; the workbook calculates from unrounded inputs. Guidance assigned to a future quarter retains Model status because guidance is not an actual result.

8. Multiple, shares and recognition convention

20×forward adjusted OP is retained for qualified specialty processes and a broadening differentiated portfolio. The multiple is applied to derived consolidated OP; it does not assume all revenue earns photonics margins. It remains a price target judgment rather than an after-tax earnings or enterprise-value calculation.

556M guided fully diluted shares anchor Q3,557M follows in Q4, then 562/568/574/580M in 2027-30. The 6M annual additions are an explicit net award/acquisition allowance rather than a reported future share forecast. No unexecuted repurchase benefit is assumed. Q2 cash/equivalents/marketable securitiesabout 3.3B versus debt 1.1B supports investment capacity, but customer prepayments and grants carry obligations. Cash, debt and exercise proceeds are not separately capitalized in this framework OP price target.

The valuation denominator is the price target quarter’s diluted-share proxy. It is not a historical EPS weighted-average assumption disguised as a fully known future capitalization. Unit factor: 1; this converts the model units to USD per share. No cash balance, enterprise-value adjustment or after-tax earnings substitution is added to the operating-profit price target.

9. Complete rolling four-quarter price target derivation

Price target quarter	Exact earnings window	Forward 4Q OP	Multiple	Diluted shares M	Price target USD	Key inflection
Q3 2026	Q3 2026 to Q2 2027	1,428.325000	20x	556.000000	51.378597	Q3 revenue/GM/expense midpoint
Q4 2026	Q4 2026 to Q3 2027	1,551.000000	20x	557.000000	55.691203	Quarterly shipment recovery
Q1 2027	Q1 2027 to Q4 2027	1,680.400000	20x	562.000000	59.800712	2027 pricing and MPS launch
Q2 2027	Q2 2027 to Q1 2028	1,823.700000	20x	562.000000	64.900356	NPO qualification conversion
Q3 2027	Q3 2027 to Q2 2028	1,974.506250	20x	562.000000	70.267126	Higher optical content; interposer qualification cost
Q4 2027	Q4 2027 to Q3 2028	2,135.775000	20x	562.000000	76.006228	2027 qualified capacity; interposer qualification cost
Q1 2028	Q1 2028 to Q4 2028	2,303.050000	20x	568.000000	81.093310	2028 SiPho ramp; pre-volume interposer phase
Q2 2028	Q2 2028 to Q1 2029	2,480.212500	20x	568.000000	87.331426	Optical yield/loading; interposer sales after overlap
Q3 2028	Q3 2028 to Q2 2029	2,649.006250	20x	568.000000	93.274868	CPO adoption scenario; interposer sales after overlap
Q4 2028	Q4 2028 to Q3 2029	2,802.987500	20x	568.000000	98.696743	SiPho 1.08B exit annualized; interposer sales after overlap
Q1 2029	Q1 2029 to Q4 2029	2,946.500000	20x	574.000000	102.665505	2029 broader optical demand; interposer sales after overlap
Q2 2029	Q2 2029 to Q1 2030	3,088.487500	20x	574.000000	107.612805	More qualified output; interposer sales after overlap
Q3 2029	Q3 2029 to Q2 2030	3,245.037500	20x	574.000000	113.067509	Power and photonics mix; interposer sales after overlap
Q4 2029	Q4 2029 to Q3 2030	3,423.337500	20x	574.000000	119.280052	2029 customer scale; interposer sales after overlap
Q1 2030	Q1 2030 to Q4 2030	3,625.900000	20x	580.000000	125.031034	2030 SiPho progression; interposer sales after overlap

- **Q3 2026:** (314.925000 + 348.600000 + 364.300000 + 400.500000) x 20 / 556.000000 = **51.378597 USD/share**. Window: Q3 2026 to Q2 2027.

- **Q4 2026:** (348.600000 + 364.300000 + 400.500000 + 437.600000) x 20 / 557.000000 = **55.691203 USD/share**. Window: Q4 2026 to Q3 2027.
- **Q1 2027:** (364.300000 + 400.500000 + 437.600000 + 478.000000) x 20 / 562.000000 = **59.800712 USD/share**. Window: Q1 2027 to Q4 2027.
- **Q2 2027:** (400.500000 + 437.600000 + 478.000000 + 507.600000) x 20 / 562.000000 = **64.900356 USD/share**. Window: Q2 2027 to Q1 2028.
- **Q3 2027:** (437.600000 + 478.000000 + 507.600000 + 551.306250) x 20 / 562.000000 = **70.267126 USD/share**. Window: Q3 2027 to Q2 2028.
- **Q4 2027:** (478.000000 + 507.600000 + 551.306250 + 598.868750) x 20 / 562.000000 = **76.006228 USD/share**. Window: Q4 2027 to Q3 2028.
- **Q1 2028:** (507.600000 + 551.306250 + 598.868750 + 645.275000) x 20 / 568.000000 = **81.093310 USD/share**. Window: Q1 2028 to Q4 2028.
- **Q2 2028:** (551.306250 + 598.868750 + 645.275000 + 684.762500) x 20 / 568.000000 = **87.331426 USD/share**. Window: Q2 2028 to Q1 2029.
- **Q3 2028:** (598.868750 + 645.275000 + 684.762500 + 720.100000) x 20 / 568.000000 = **93.274868 USD/share**. Window: Q3 2028 to Q2 2029.
- **Q4 2028:** (645.275000 + 684.762500 + 720.100000 + 752.850000) x 20 / 568.000000 = **98.696743 USD/share**. Window: Q4 2028 to Q3 2029.
- **Q1 2029:** (684.762500 + 720.100000 + 752.850000 + 788.787500) x 20 / 574.000000 = **102.665505 USD/share**. Window: Q1 2029 to Q4 2029.
- **Q2 2029:** (720.100000 + 752.850000 + 788.787500 + 826.750000) x 20 / 574.000000 = **107.612805 USD/share**. Window: Q2 2029 to Q1 2030.
- **Q3 2029:** (752.850000 + 788.787500 + 826.750000 + 876.650000) x 20 / 574.000000 = **113.067509 USD/share**. Window: Q3 2029 to Q2 2030.
- **Q4 2029:** (788.787500 + 826.750000 + 876.650000 + 931.150000) x 20 / 574.000000 = **119.280052 USD/share**. Window: Q4 2029 to Q3 2030.
- **Q1 2030:** (826.750000 + 876.650000 + 931.150000 + 991.350000) x 20 / 580.000000 = **125.031034 USD/share**. Window: Q1 2030 to Q4 2030.

10. Annual trajectory cross-check

Calendar year	Revenue	GP / contribution	Opex	OP	OP margin
2027	8,295.000	2,865.400	1,185.000	1,680.400	20.258%
2028	9,685.000	3,687.300	1,384.250	2,303.050	23.780%
2029	11,070.000	4,515.000	1,568.500	2,946.500	26.617%
2030	12,550.000	5,354.400	1,728.500	3,625.900	28.892%

These are calendar-year sums of the quarterly path, not extra revenue and not the live valuation method. For non-December fiscal reporters, fiscal guidance and these calendar totals must be aligned using the period mapping above; a four-quarter run rate is not automatically the same fiscal year. Outer-year outputs are conditional TAM/capture trajectories, not verified management sales targets.

Company annual sales sum $8,295M/$9,685M/$11,070M/$12,550M in 2027-30, including the incremental interposer branch once. SiPho is a growing subset, not the whole CID category. Q4 2028 sales 270M×4=1.08B and Q4 2030 540M×4=2.16B; annual totals are 900M and 1.95B respectively. A2B exit rate cannot be placed in every 2030 quarter without accelerating the production schedule beyond the cited framing.

11. Historical Q3 2027 dilution diagnostic

This table is the September 17 snapshot and retains its stated Q3 2027 window and pre-interposer operating case. The current active Q4 2027 target is $76.006228; the older diagnostic is not an economic downgrade or an alternative active horizon.

Denominator sensitivity	Q3 2027 shares M	Q3 2027 price target USD	Effect on price target
-5% shares	533.900000	74.053193	+5.26%
+0% shares	562.000000	70.350534	+0.00%
+5% shares	590.100000	67.000508	-4.76%
+10% shares	618.200000	63.955031	-9.09%

This isolates denominator uncertainty; it is not a second bull/bear operating case or permission to remove known dilution. The current base case remains the explicit share path in the rolling table; the historical diagnostic is not recalculated as a new case.

12. Programme and probability ledger

Programme / signal	Source and probability	Amount / unit base	Treatment in this model	Escalation / recognition
Existing optical production	Reported end-market growth;95% continuation confidence	CID actuals and qualified processes	Calibrates the initial allocation	Actual SiPho/SiGe disclosure would replace analytical splits.
Marvell SiGe agreement	Named customer partnership;85% execution confidence	Undisclosed volumes/pricing	Inside other CID; no added agreement value	Qualified Vermont capacity and customer shipments.
MPS production	Signed partnership;80% timing confidence	Early 2027 start; undisclosed revenue	Data-center portion inside CID, remaining applications in other markets	Process transfer, yield and volume start.
SCALE NPO/CPO	Customer tape-outs;70% conversion confidence	No complete public unit schedule	Inside SiPho trajectory, not another whole market	Customer qualification and accepted volume sales.
TSMC interposer supply	Signed agreement; 85% execution scenario	$2B headline, initial five-year term; H1 2028 expected ramp	Explicit accepted-sales schedule with 50% overlap and full incremental cost bridge	Malta tooling, qualified output, acceptance, capex/payment terms and realized margin.
Quantum support	Finalized milestone award;85% funding confidence	Up to 375M overfiveyears	No immediate sales/OP windfall	Grant accounting and milestones must be reconciled separately.
Photonics funding	Letter of intent;60% realization confidence	300M proposal	No modeled OP uplift	Definitive agreement and recognized accounting treatment.

Probabilities are analyst judgments under the framework signal hierarchy. Embedded revenue is not multiplied or added a second time. Where no defensible amount or unit base exists, the opportunity remains explicitly unquantified; confidence alone cannot create a revenue number. New evidence can enlarge the TAM, improve capture or move recognition earlier, but each requires an identifiable model input.

13. Scored risk classes

Risk class	Scored deduction	Mechanism and consequence
Capex and qualification	-1	Tools, yield and customer qualification determine usable capacity and cash returns.
Customer concentration	-1	A few large customers and programme changes can move whole technology corridors.
Competition and legacy cycle	-1	Mobile/auto softness and competing foundries can offset optical growth.
Adjusted economics and dilution	-1	Recurring SBC is excluded; acquisitions and future awards dilute common claims.
Policy and supply chain	-1	Geopolitics, funding milestones and critical materials can change cost, timing and demand.

14. Dated catalysts, counters and escalation

1. **2026-11-15 monitoring date - Q3 print** Test 1,885M revenue,30.5% GM and 260M expenses.
2. **2027-03-31 - MPS initial production** Confirm process qualification and recognized shipments; a partnership announcement is insufficient.
3. **2027-12-31 - SiGe and NPO** Track usable capacity, pricing and converted tape-outs.
4. **2028-12-31 - SiPho exit milestone** Compare actual quarter with 270M model; distinguish annual revenue from annualized run rate.
5. **2030-12-31 - Larger photonics case** 540M exit quarter requires funded, qualified output and customer capture.
6. **Every report - Costs and shares** Check manufacturing cost, R&D, SBC and net dilution against the path.

15. Cross-book monitoring

Marvell is a verified SiGe customer through the September 17 capacity agreement. IQE/Tower and GF offer competing or complementary technology routes, not assured simultaneous capture of every optical link. Foundry revenue and downstream component sales represent different invoices within the same end-demand pool. No unverified exclusive AMD device attribution or fixed Sivers-to-GF capacity multiplier is used. MRVL's use of GF SiGe does not alone establish a loss at Tower: product process, customer socket and sourcing share must be identified. IQE epi content and external laser suppliers may serve platform-specific designs without growing in a universal fixed ratio to GF revenue.

16. Source register and model limits

G8 - TSMC interposer manufacturing agreement (2026-10-08). Primary issuer announcement. Agreement existence is verified; timing, acceptance, payment mix and realized margin remain uncertain. The October 11 Grok relay and the original announcement are one economic event.

G1 - Q2 results and Q3 guidance (2026-08-05). Reported financials and Q3 midpoint anchor the first forecast quarter. SBC remains excluded from the chosen non-IFRS basis.

G2 - Q2 management call (2026-08-05). CID was 16% of Q2 revenue, +62% year on year. SiPho is expected to more than double in 2026; SiGe capacity is oversubscribed through 2027. NPO/CPO ramps were discussed for 2027/2028. Technology-services revenue is already inside end-market totals.

G3 - Goldman Sachs management discussion (2026-09-08). The discussion refers to SiPho USD 1B exit-2028 and USD 2B 2030 run-rate milestones. Capacity build and qualification constrain near-term conversion. These are not full-year revenue figures.

G4 - Expanded Marvell SiGe agreement (2026-09-17). Multi-year agreement expands Vermont SiGe capacity for Marvell optical requirements. No contract revenue, wafer volume or price is disclosed.

G5 - MPS manufacturing partnership (2026-09-09). MPS proprietary processes will be deployed in Singapore, with volume production expected in early 2027. No revenue amount is stated.

G6 - Quantum funding agreement (2026-09-08). Up to USD 375M over five years, conditional on milestones. The separate USD 300M photonics program remains an LOI in this release; neither is booked as an immediate earnings windfall.

G7 - Prior-quarter management transcript (2026-05-05). Cross-check confirms the prior SiPho exit-run-rate framing and SiGe capacity constraints.

Product-level revenue and margins, outer-year growth and share trajectories are assumptions. No complete two-call Q&A review. The framework values consolidated adjusted OP without a separate cash/debt bridge, taxes or SBC expense; it is a conditional price target rather than a fair-value estimate.

Workbook: v3_2_master_portfolio.xlsx, model tab GFS_Q_Model. Portfolio ranking, prices and return calculations stay in the master workbook. **October 11 update ID:** GROK-20261011-2105-GFS-INTERPOSERS. Grok XDATA-20261011-2003-ET-BQYV R0, source posts 2109355351099556085 and 21094272154108523 27/2109427277125853186, reconciled to the October 8 issuer event.

Current artifact: G:/My Drive/1amInvesting/Artifacts/GFS_GlobalFoundries_DeepDive_Q2_2026.pdf, with synchronized self-contained Markdown. **Original snapshot:** versions/originals/GFS_GlobalFoundries_DeepDive_Q2_2026.md. Superseded history is retained separately in VERSION_LOG, not in this current-state model.